

Initial and subject parameter distribution:

$$\underbrace{\begin{bmatrix} \text{auto_posaff_W}_i \\ \text{cint_posaff_B}_i \\ \text{T0MEANS_posaff_B}_i \\ \text{T0MEANS_posaff_W}_i \end{bmatrix}}_{\phi(i)} \approx \text{N} \left(\begin{bmatrix} -5.257 \\ 0.004 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 3.765 & 0.006 & 0 & 0 \\ 0.006 & 0.006 & 0 & 0 \\ 0 & 0 & 0.28 & 0 \\ 0 & 0 & 0 & 0.152 \end{bmatrix} \right) + \underbrace{\begin{bmatrix} -0.242 & -0.525 & 0.593 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}}_{\hat{\beta}} \underbrace{\begin{bmatrix} \text{zpos} \\ \text{zneg} \\ \text{zdis} \end{bmatrix}}_{\mathbf{z}}$$

Deterministic change:

$$\text{d} \underbrace{\begin{bmatrix} \text{posaff_B} \\ \text{posaff_W} \end{bmatrix}}_{\text{d}\boldsymbol{\eta}(t)}(t) = \left(\underbrace{\begin{bmatrix} -0.023 & 0 \\ 0 & -5.257 \end{bmatrix}}_{\mathbf{A}_{\text{DRIFT}}} \underbrace{\begin{bmatrix} \text{posaff_B} \\ \text{posaff_W} \end{bmatrix}}_{\boldsymbol{\eta}(t)}(t) + \underbrace{\begin{bmatrix} 0.004 \\ 0 \end{bmatrix}}_{\mathbf{b}_{\text{CINT}}} \right) dt +$$

Random change:

$$\underbrace{UcorSDtoChol \left\{ \begin{bmatrix} 0 & 0 \\ 0 & 1.8 \end{bmatrix} \right\}}_{\mathbf{G}_{\text{DIFFUSION}}} \text{d} \underbrace{\begin{bmatrix} W_1 \\ W_2 \end{bmatrix}}_{\text{d}\mathbf{W}(t)}(t)$$

Observations:

$$\underbrace{[\text{esm_pa}]}_{\mathbf{Y}(t)}(t) = \underbrace{\begin{bmatrix} 1 & 1 \end{bmatrix}}_{\mathbf{\Lambda}_{\text{LAMBDA}}} \underbrace{\begin{bmatrix} \text{posaff_B} \\ \text{posaff_W} \end{bmatrix}}_{\boldsymbol{\eta}(t)}(t) + \underbrace{\begin{bmatrix} 0 \end{bmatrix}}_{\boldsymbol{\tau}_{\text{MANIFESTMEANS}}} +$$

Observation noise:

$$\underbrace{UcorSDtoChol \left\{ [0.585] \right\}}_{\boldsymbol{\Theta}_{\text{MANIFESTVAR}}} \underbrace{[\epsilon_1]}_{\boldsymbol{\epsilon}(t)}(t)$$

System noise
distribution per time step: $\Delta[W_{j \in [1,2]}](t-u) \sim \text{N}(0, t-u)$

Note: *UcorSDtoChol* converts lower tri matrix of standard deviations and unconstrained correlations to Cholesky factor, *UcorSDtoCov* = transposed cross product of *UcorSDtoChol*, to give covariance, See Driver & Voelkle (2018) p11.
Individual specific notation (subscript i) only shown for subject parameter distribution – pop. means shown elsewhere.
Linearised approximation of subject parameter distribution shown.